Answer key: ECON 2 Practice final B

Name: _____ SID: _____ · 90 minutes · 60 points · one page of notes, front and back · calculator allowed

The exam

A workable split is about 15 minutes on Question 1, 15 on Question 2, 35 on Question 3, and 20 on Question 4, which leaves 5 minutes to check your work. Edwards tells you to confine answers to the box and that less is more, so practice saying each idea in two to four sentences. When you grade a part, compare it to the "Full credit" line, which is modeled on what his Fall 2025 key wrote in red.

Abbreviations used below: PAE is planned aggregate expenditure (C + I + G + NX), mpc is the marginal propensity to consume, GDP is gross domestic product, MPL and MPK are the marginal products of labor and capital, MR is marginal revenue, and MRPL and MRPK are the marginal revenue products of labor and capital (MR × MPL and MR × MPK).

Admin (2 points)

1 point Write your name and student ID at the top of each page of the exam.

1 point If the Fed let you set one interest rate for a single day, which rate would you pick, and what would you set it to? (There are no wrong answers.)

1. (12 points total) Multiple choice: the margin of victory

For each of these questions, circle the one correct answer.

2 points 1(a). You split your daily snack budget between coffee and frozen yogurt. You are maximizing your utility when

(i) The marginal utility of coffee equals the marginal utility of frozen yogurt
(ii) The marginal utility per dollar spent is the same for coffee and frozen yogurt
(iii) You buy the same number of coffees and frozen yogurts
(iv) The total utility from coffee equals the total utility from frozen yogurt

2 points 1(b). At the quantity a profit-maximizing monopolist chooses to sell,

(i) Marginal benefit equals marginal cost, so the outcome is efficient
(ii) Marginal benefit is greater than marginal cost, so producing more would add more to society's benefit than to its cost
(iii) Marginal benefit is less than marginal cost, so the monopolist produces too much
(iv) Price equals marginal revenue

2 points 1(c). Which of these shifts the labor demand curve for hospital nurses to the right?

(i) New monitoring equipment lets each nurse care for more patients per shift
(ii) Hospitals can charge more for each patient visit
(iii) The wage of nurses falls
(iv) Both (i) and (ii)

2 points 1(d). Over long periods of time, the key source of rising real GDP per person is

(i) Higher average labor productivity, which comes mainly from capital accumulation and technological progress
(ii) Larger government budget deficits
(iii) Lower interest rates set by the Federal Reserve
(iv) A weaker currency that raises net exports

2 points 1(e). Suppose inflation rises well above the Fed's 2% target. According to the Fed's reaction function, the Fed will

(i) Raise the real interest rate, which lowers planned spending and output and over time brings inflation down
(ii) Lower the real interest rate so that output rises and prices fall
(iii) Raise government purchases to absorb the extra spending
(iv) Do nothing, because inflation always returns to target on its own

2 points 1(f). If a country's domestic saving is smaller than its domestic investment, then

(i) It has net capital inflows and runs a trade deficit
(ii) It has net capital outflows and runs a trade surplus
(iii) Its trade must be balanced
(iv) Its currency must appreciate every year

2. (10 points total) Short answers, real returns

Answer each part in the box in a few sentences. Show your work where there is arithmetic.

2 points 2(a). Which of these transactions this year add to this year's U.S. GDP? For each one, say yes or no and give the reason in a few words. (1) A student buys a new laptop that was assembled this year in a factory in Fremont. (2) A tire company in Ohio sells $40,000 of tires to an automaker, which puts them on new cars that it sells this year. (3) A family buys a house built in 1995 directly from its previous owners. (4) A student buys a new phone that was assembled this year in China.

2 points 2(b). A tiny economy produces only burritos and textbooks. In year 1 it makes 100 burritos at $10 each and 20 textbooks at $50 each. In year 2 it makes 110 burritos at $12 each and 20 textbooks at $60 each. Find nominal GDP in both years and real GDP in year 2 using year 1 as the base year. By what percent did nominal GDP and real GDP grow? Show your work.

2 points 2(c). The exchange rate moves from 20 Mexican pesos per U.S. dollar to 16 pesos per U.S. dollar. Which currency appreciated and which depreciated? A hotel in Mexico City charges 800 pesos a night. What does a night cost an American traveler in dollars before and after? What happens to U.S. net exports to Mexico?

2 points 2(d). In one day, a typical U.S. worker can produce either 4 tons of oranges or 8 pairs of sneakers. A typical Vietnamese worker can produce either 1 ton of oranges or 6 pairs of sneakers. Which country has an absolute advantage in each good? Which has a comparative advantage in each good? What should each country export? Show your work.

2 points 2(e). A small town has 10 households with these annual incomes, in thousands of dollars: 20, 30, 40, 50, 60, 70, 80, 100, 250, and 300. What share of total income goes to the top 10% of households? Researchers like Piketty and Saez measure top income shares using income tax returns rather than household surveys. Why?

3. (24 points total) Shaken confidence: a Keynesian cross-examination

Suppose that at the start of 2027, potential output is Y* = $30 trillion, the marginal propensity to consume is mpc = 0.6, and this system models planned aggregate expenditures (PAE) in trillions of dollars:

Component	Equation
Consumption	C = 6 + 0.6Y
Planned investment	I = 4
Government purchases	G = 4
Net exports	NX = −2 (a trade deficit of $2 trillion)
Planned aggregate expenditure	PAE = C + I + G + NX

2 points 3(a). Combine the expenditure terms in PAE and find the level of PAE when Y = 0, which is the vertical intercept in the Keynesian cross diagram. Show your work.

2 points 3(b). Using your answer to part (a), set PAE = Y and solve for the short-run equilibrium level of Y. Is there an output gap? Show your work.

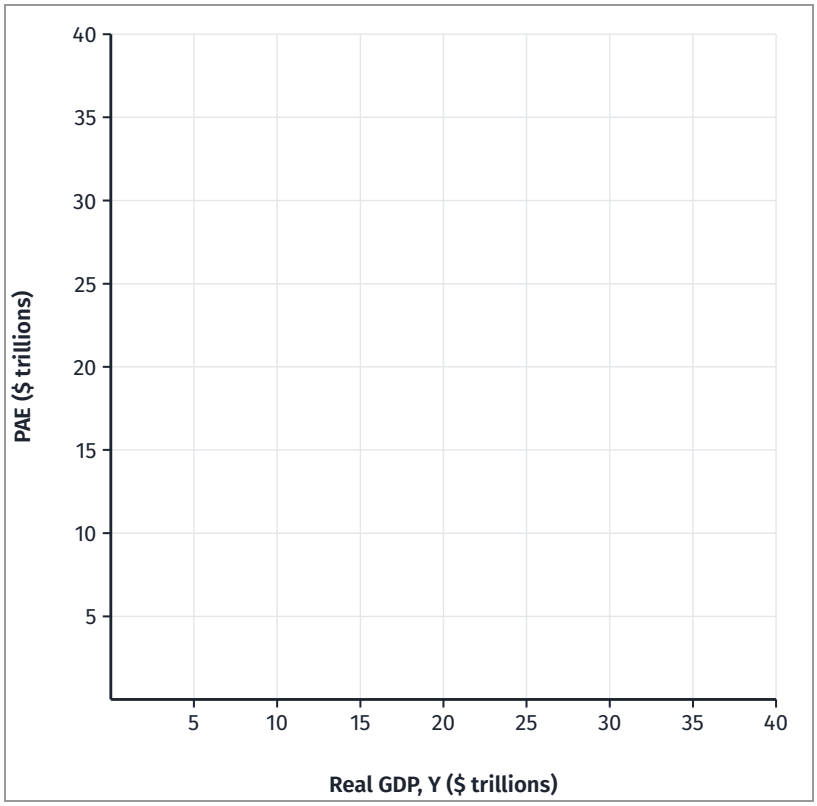
In early 2027, a survey shows consumer confidence falling as tariffs raise the prices of imported goods, and households cut back on spending at every level of income. At the same time, trading partners retaliate with tariffs on U.S. exports. Suppose retaliation cuts U.S. exports by more than the U.S. tariffs cut imports, so the new situation is:

Component	Equation
Consumption	C' = 5.4 + 0.6Y
Planned investment	I = 4
Government purchases	G = 4
Net exports	NX' = −2.6
Planned aggregate expenditure	PAE' = C' + I + G + NX'

2 points 3(c). Combine the expenditure terms in PAE' and find its vertical intercept. Show your work.

2 points 3(d). Set PAE' = Y and solve for the new short-run equilibrium Y. What is the output gap now, and what kind of gap is it? Show your work.

4 points 3(e). On the axes below, graph the original PAE line from part (a), the new PAE' line from part (c), and the 45-degree line. Show both equilibria and label the intercepts, the lines, and the values on both axes.



Blank axes, as on the exam. Both axes use the same scale, so your 45-degree line should really be at 45 degrees.

2 points 3(f). By how much did planned aggregate expenditure change between (a) and (c)? Express it in trillions of dollars and say whether it is an increase, a decrease, or no change. Explain fully what the arithmetic means about planned spending and how it appears in the graph.

2 points 3(g). By how much did GDP change between (b) and (d)? Express it in trillions of dollars and say whether it is an increase, a decrease, or no change. Explain fully what the arithmetic means about GDP and how it appears in the graph.

2 points 3(h). Are your answers to (f) and (g) the same or different? If different, which is larger in magnitude, and why? Compute the multiplier and use it to connect the two numbers. Explain fully, referencing the model.

Congress is gridlocked, so the Federal Reserve decides to respond. Before the shock, the Fed's policy rate (the nominal interest rate it sets) is 6% and inflation is 2%.

2 points 3(i). Explain fully how the Fed cutting its policy rate raises planned investment and consumption. Then show on your graph from (e) what the cut does to the PAE line and to output.

2 points 3(j). Suppose each 1 percentage point cut in the real interest rate raises planned investment by $0.3 trillion and autonomous consumption by $0.1 trillion, and inflation stays at 2% in the short run. Ignore any effect on net exports. How much autonomous spending is needed to close the gap from (d), and what policy rate should the Fed set? Show your work.

2 points 3(k). Suppose the Fed overshoots and cuts its policy rate by 4 percentage points instead. What is the new equilibrium output, and what does the model say will happen to inflation and to the Fed's rate after that? Explain.

4. (12 points total) Packing it in: machines, workers, and present value

A fruit packing house in the Central Valley is deciding whether to buy a new sorting and packing line, which costs $14,000 and lasts two years. The line needs a crew of workers to run it. The packing house sells in a competitive market at $2 per box, so its marginal revenue is MR = $2 per box. The line's marginal product of capital (MPK) is 5,500 extra boxes in year 1 and 3,025 extra boxes in year 2, as the machine wears out. Payments arrive at the end of each year. Recall that the present value of a stream of payments F_t is

$$PV = F_1 \div (1 + i) + F_2 \div (1 + i)^2$$

and that for a unit of capital, F_t is its marginal revenue product, $MRPK(t) = MR(t) \times MPK(t)$.

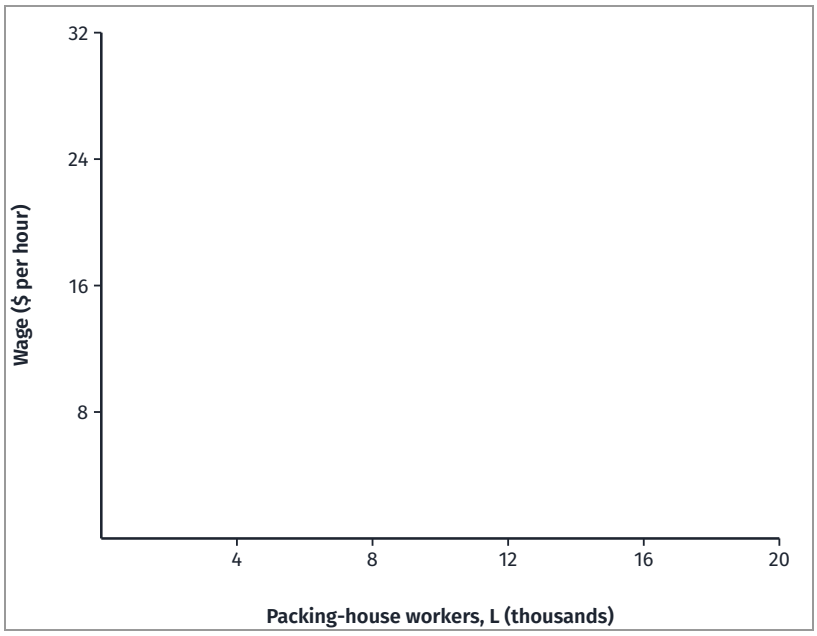
2 points 4(a). Compute the line's MRPK in each year and its present value when the interest rate is i = 10%. Should the packing house buy it? Show your work.

2 points 4(b). Suppose the interest rate rises to i = 20%. Recompute the present value. What happens to the price of a unit of this kind of capital, and does the packing house still buy the line? Explain.

2 points 4(c). Go back to i = 10%. Now suppose stepped-up immigration enforcement is expected to remove many of the workers who staff packing lines, so in year 2 the line can only be run part time and its MPK falls to 1,815 boxes. What happens to the line's present value and to the price of this kind of capital? Should the packing house still buy it? Show your work and explain.

Now consider the labor market for packing-house workers, holding the number of packing machines fixed. Recall that labor demand is the marginal revenue product of labor, MRPL = MR × MPL, and labor supply reflects workers' marginal cost of supplying their time. Suppose labor demand is w = 30 − L and labor supply before enforcement is w = 10 + L, where w is the hourly wage and L is thousands of workers. Enforcement removes 4 thousand workers at every wage.

4 points 4(d). Illustrate the labor market before and after the enforcement. Show all demand and supply curves, and the equilibrium wages and employment with values. Briefly explain.



Blank axes, as on the exam.

2 points 4(e). Who wins and who loses in this market, given what you found in (c) and (d)? Why is the MRPL of the last worker hired now higher than before?

Score it

Grade yourself part by part against the "Full credit" lines, and give yourself half credit when you had the right idea but a wrong number or a missing graph label. Then reread the unit listed for every part you missed.

Part	Points	What it tests	Reread
Admin	2	Name and SID, fun question	00 How the final works
1(a)	2	Rational spending rule	13 Midterm material refresher
1(b)	2	Monopoly: MB above MC	13 Midterm material refresher

1(c)	2	What shifts labor demand	01 Labor and wages
1(d)	2	Source of long-run growth	07 Long-run growth
1(e)	2	The Fed's reaction function	11 Inflation and the return to potential
1(f)	2	Saving, investment, and the trade deficit	12 Exchange rates and international macro
2(a)	2	What counts in GDP	06 GDP, prices, and unemployment
2(b)	2	Nominal versus real GDP	06 GDP, prices, and unemployment
2(c)	2	Exchange rate quote, appreciation, net exports	12 Exchange rates and international macro
2(d)	2	Comparative advantage between countries	04 Gains from international trade
2(e)	2	Top income shares and tax data	02 Inequality
3(a) to 3(e)	12	Building PAE, solving PAE = Y, output gap, the to-scale graph	08 Recessions and the Keynesian cross
3(f) to 3(h)	6	Change in PAE versus change in GDP, the multiplier	08 Recessions and the Keynesian cross and 09 Fiscal policy
3(i), 3(j)	4	How a rate cut raises I and C, sizing the cut	10 Monetary policy and financial markets
3(k)	2	Overshooting, inflation, and the reaction function	11 Inflation and the return to potential
4(a) to 4(c)	6	MRPK, present value, the price of capital	03 Capital, interest, and present value
4(d), 4(e)	6	Labor market shift, winners and losers	01 Labor and wages
Total	**60**		

The three habits this exam drills: the change in GDP equals the multiplier, 1 ÷ (1 − mpc), times the vertical shift in PAE, so a policy only has to undo the shift in PAE. A rate cut works because inflation moves slowly, so a lower nominal rate means a lower real rate, which raises planned investment and interest-sensitive consumption. The price of capital is the present value of its future MRPKs, so it falls when interest rates rise or when anything, like less future labor, lowers its future marginal product.